Microsoft

PostgreSQL and GraphRAG Integration in Docker

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Microsoft

Background and objectives:

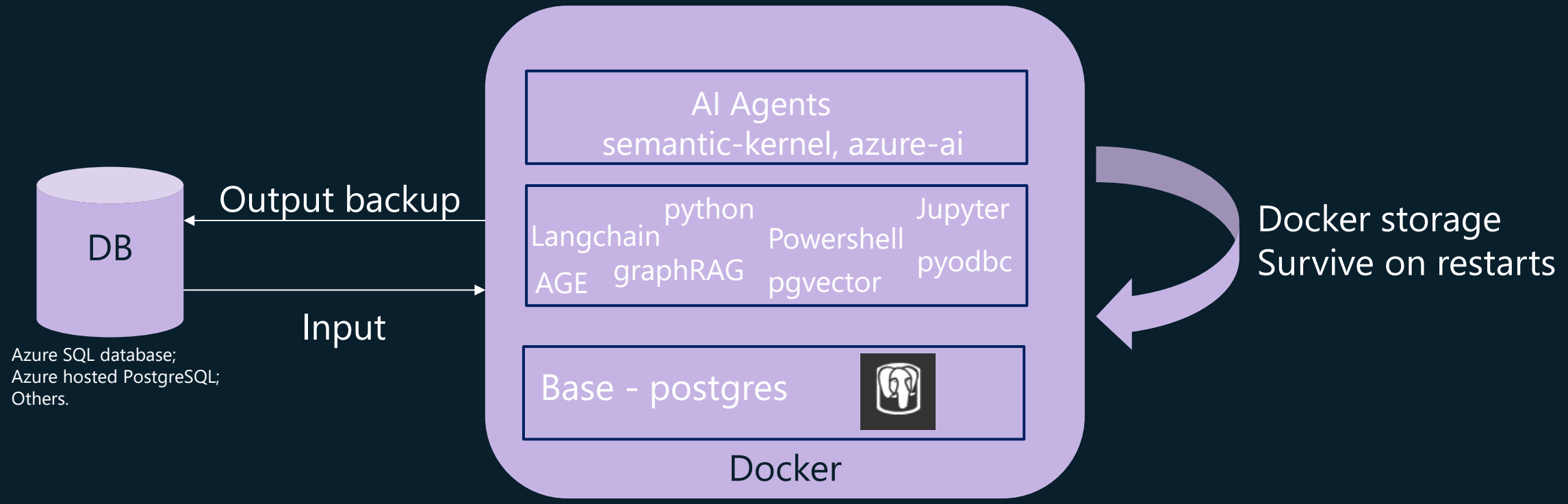
- Raw input data is already in a DB.
- Open Cypher query capability is needed.
- Run CLI version of graphRAG Conveniently.
- Avoid another blob storage between the DB and graphRAG.

Solution design:

- Build graphARG index directly using data in the DB.
- Integrate PostgreSQL and graphRAG in a Docker container, add Apache AGE for running Cypher query.
- Bonus: add AI agent services on top.

Architecture

The docker image uses postgres as base, then added python, graphRAG, AGE, semantic-kernel and other needed packages. Two volumes are created to persist postgres data and app related data.




```yaml
volumes:
  - local_postgres_data:/var/lib/postgresql/data
  - graphrag_data:/app/graphrag-folder
```

Preparation

1. Prepare Dockerfile
2. Prepare the .env file
3. Prepare setting.yaml
4. Prepare docker-compose.yaml
5. Prepare python code or jupyter notebook

.env example

Your DB type,
you can define yours

Your DB properties

Your docker Postgres property

If you use your docker
Postgres, then set to true

```
DB_TYPE="Azure SQL"  # "Azure SQL" or "Azure postgres"
MY_DB_HOST=
MY_DB_PORT=
MY_DB_USER=
MY_DB_PASSWORD=
MY_DB_NAME=
MY_DB_SSLMODE=disable
PROMPTS=/app/graphrag-folder/prompts
CONFIG_PATH=/app/graphrag-folder/settings.yaml
AGE_HOST=postgres
AGE_PORT=5432
AGE_USER=postgres
AGE_PASSWORD=xxxx
AGE_DB=postgres
USE_LOCAL_AGE=true
AOAI_API_BASE="https://xxxx.openai.azure.com"
AOAI_API_KEY=xxxx
AOAI_LLM_MODEL=gpt-4o
AOAI_LLM_API_VERSION="2025-01-01-preview"
AOAI_LLM_DEPLOYMENT=gpt-4o
AOAI_EMBEDDING_MODEL=text-embedding-3-small
AOAI_EMBEDDING_API_VERSION="2023-05-15"
AOAI_EMBEDDING_DEPLOYMENT=text-embedding-3-small
```

Step 0 – insert .txt input to DB if not already existing

Run python code to insert .txt files to DB:

➢ python insert-table.py

The code insert-table.py provides two methods of DB connections:

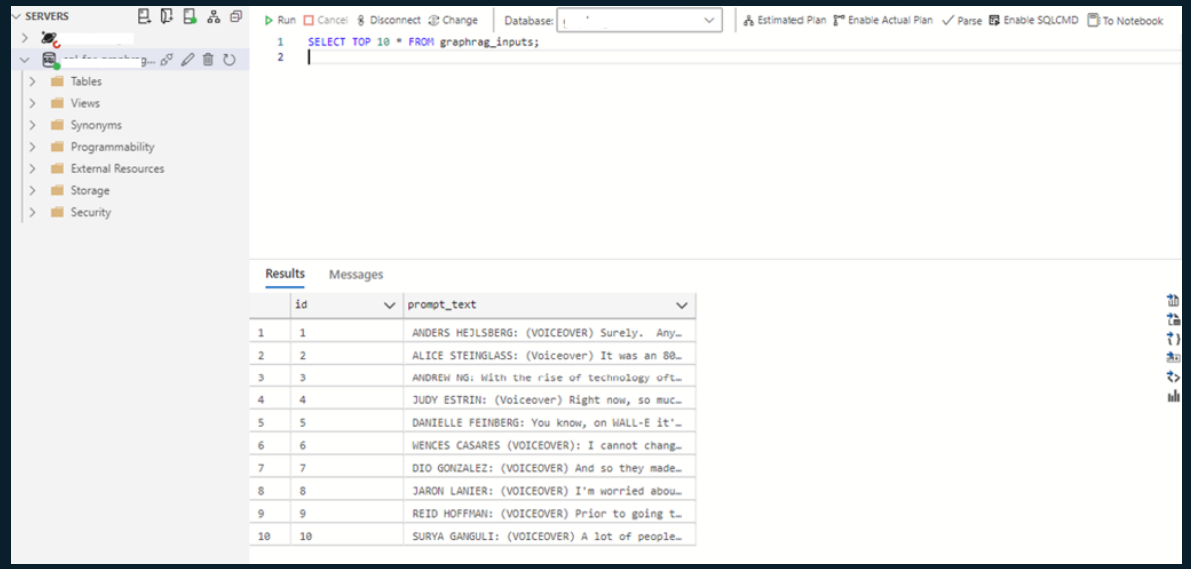
- For inserting table to Azure hosted PostgreSQL.
- Or for inserting table to Azure SQL database.

Below is example of inserting to Azure SQL database:

> Python insert-table.py

```
['SQL Server', 'Microsoft Access Driver (*.mdb, *.accdb)', 'Microsoft Excel Driver (*.xls, *.xlsx, *.xlsm, *.xlsb)', 'Microsoft Access Text Driver (*.txt, *.csv)',
'Microsoft Access dBASE Driver (*.dbf, *.ndx, *.mdx)', 'ODBC Driver 18 for SQL Server']
Processed file: EP-01_Behind_the_Tech_Kevin_Scott_Podcast_with_Anders_Hejlsberg.txt
Number of lines in EP-01_Behind_the_Tech_Kevin_Scott_Podcast_with_Anders_Hejlsberg.txt: 241
Processed file: EP-02_Behind_the_Tech_Kevin_Scott_Podcast_with_Alice_Steinglass.txt
Number of lines in EP-02_Behind_the_Tech_Kevin_Scott_Podcast_with_Alice_Steinglass.txt: 207
Processed file: EP-03_Behind_the_Tech_Kevin_Scott_Podcast_with_Andrew_Ng.txt
Number of lines in EP-03_Behind_the_Tech_Kevin_Scott_Podcast_with_Andrew_Ng.txt: 159
Processed file: EP-04_Behind_the_Tech_Kevin_Scott_Podcast_with_Judy_Estrin.txt
Number of lines in EP-04_Behind_the_Tech_Kevin_Scott_Podcast_with_Judy_Estrin.txt: 200
Processed file: EP-05_Behind_the_Tech_Kevin_Scott_Podcast_with_Danielle_Feinberg.txt
Number of lines in EP-05_Behind_the_Tech_Kevin_Scott_Podcast_with_Danielle_Feinberg.txt: 291
Processed file: EP-06_Behind_the_Tech_Kevin_Scott_Podcast_with_Wences_Casares.txt
Number of lines in EP-06_Behind_the_Tech_Kevin_Scott_Podcast_with_Wences_Casares.txt: 265
Processed file: EP-07_Behind_the_Tech_Kevin_Scott_Podcast_with_Dio_Gonzalez.txt
Number of lines in EP-07_Behind_the_Tech_Kevin_Scott_Podcast_with_Dio_Gonzalez.txt: 702
Processed file: EP-08_Behind_the_Tech_Kevin_Scott_Podcast_with_Jason_Lanier.txt
Number of lines in EP-08_Behind_the_Tech_Kevin_Scott_Podcast_with_Jason_Lanier.txt: 426
Processed file: EP-09_Behind_the_Tech_Kevin_Scott_Podcast_with_Reid_Hoffman.txt
Number of lines in EP-09_Behind_the_Tech_Kevin_Scott_Podcast_with_Reid_Hoffman.txt: 388
Processed file: EP-10_Behind_the_Tech_Kevin_Scott_Podcast_with_Surya_Ganguli.txt
Number of lines in EP-10_Behind_the_Tech_Kevin_Scott_Podcast_with_Surya_Ganguli.txt: 254
All .txt files inserted into Azure SQL Database.
```

Azure Data Studio

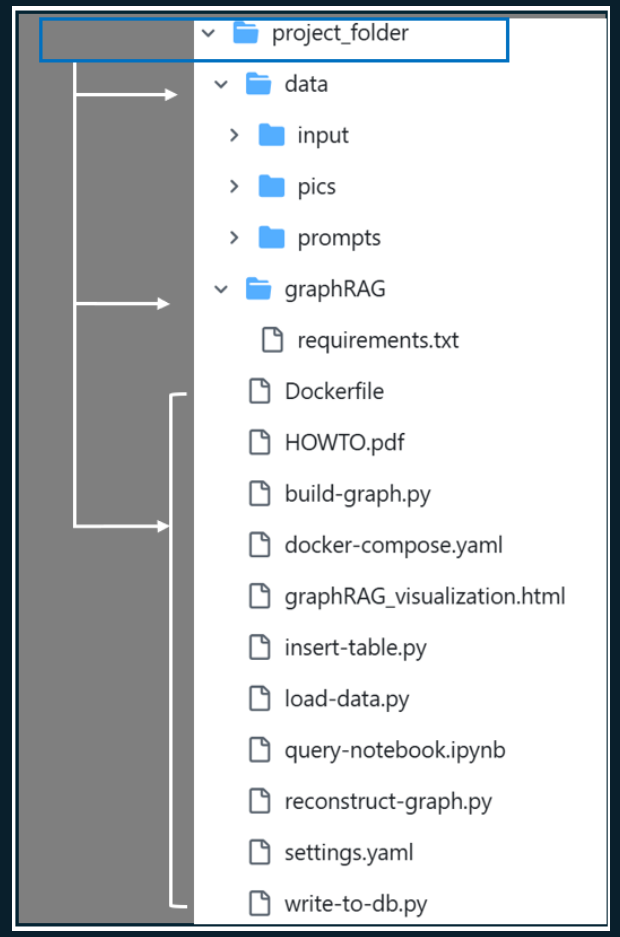


Step 1 – build docker image

This step builds docker images to include all modules needed to run services later.

In project_folder:

```
$ docker build -t graphrag-img .
```



Step 2 – run postgres service

This step starts postgres service which other services depend on.

```
$ docker compose up postgres
```

```
[+] Running 2/2
 ✓ Network postgresql-age_default  Created                                                    0.1s
 ✓ Container postgres              Created                                                    0.1s
Attaching to postgres
postgres  |
postgres  | PostgreSQL Database directory appears to contain a database; Skipping initialization
postgres  |
postgres  | 2025-05-23 16:28:27.226 UTC [1] LOG:  starting PostgreSQL 16.9 (Debian 16.9-1.pgdg120+1) on x86_64-pc-linux-gnu, compil
ed by gcc (Debian 12.2.0-14) 12.2.0, 64-bit
postgres  | 2025-05-23 16:28:27.227 UTC [1] LOG:  listening on IPv4 address "0.0.0.0", port 5432
postgres  | 2025-05-23 16:28:27.227 UTC [1] LOG:  listening on IPv6 address "::", port 5432
postgres  | 2025-05-23 16:28:27.234 UTC [1] LOG:  listening on Unix socket "/var/run/postgresql/.s.PGSQL.5432"
postgres  | 2025-05-23 16:28:27.250 UTC [29] LOG:  database system was shut down at 2025-05-23 16:05:41 UTC
postgres  | 2025-05-23 16:28:27.270 UTC [1] LOG:  database system is ready to accept connections
```

Step 3.1 – run ‘load-data’ service

This step loads data from DB to Docker folder /app/graphrag-folder/input as graphRAG input.

```
$ docker compose up load-data
```

```
[+] Running 2/2
 ✓ Container postgres       Running                                                    0.0s
 ✓ Container load-data-app  Created                                                    0.1s
Attaching to load-data-app
load-data-app  | All rows saved as .txt files in /app/graphrag-folder/input.
load-data-app exited with code 0
```

Step 3.2 – check data is present in postgres server

This sub-step checks docker folder /app/graphrag-folder/input to verify graphRAG input.

```
$ docker exec -it postgres bash
```

```
root@fffff:/app/graphrag-folder# cd input
root@fffff:/app/graphrag-folder/input# ls
input_10.txt  input_1.txt  input_2.txt  input_3.txt  input_4.txt  input_5.txt  input_6.txt  input_7.txt  input_8.txt  input_9.txt
```

Step 4.1 – build graphrag index

This step runs ‘graphrag index’ using /app/graphrag-folder/input and generate output.

```
$ docker compose up graphrag-index
```

```
[+] Running 3/3
 ✓ Container postgres            Running                                                    0.0s
 ✓ Container load-data-app       Created                                                    0.0s
 ✓ Container graphrag-index-app  Created                                                    0.1s
Attaching to graphrag-index-app
graphrag-index-app  | Total files: 10
```

```
graphrag-index-app  |
graphrag-index-app  | [119 rows x 2 columns]}
graphrag-index-app  | ⠋ GraphRAG Indexer
graphrag-index-app  | ├── Loading Input (InputFileType.text) - 10 files loaded (0 filtered) ━ 100%  0…
graphrag-index-app  | ├── create_base_text_units ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 100% 0:00:00 0:00:00
graphrag-index-app  | ├── create_final_documents ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 100% 0:00:00 0:00:00
graphrag-index-app  | ├── extract_graph ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 100% 0:00:00 0:00:00
graphrag-index-app  | ├── finalize_graph ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 100% 0:00:00 0:00:00
graphrag-index-app  | ├── create_communities ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 100% 0:00:00 0:00:00
graphrag-index-app  | ├── create_final_text_units ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 100% 0:00:00 0:00:00
graphrag-index-app  | ├── create_community_reports ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 100% 0:00:00 0:00:00
graphrag-index-app  | ├── generate_text_embeddings ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 100% 0:00:00 0:00:00🚀 All workflows completed su
ccessfully.
graphrag-index-app exited with code 0
```

Step 4.2 – check data is present in postgres server

This sub-step verifies ‘graphrag index’ output in /app/graphrag-folder/output.

```
$ docker exec -it postgres bash
```

```
root@fffff:/app/graphrag-folder# cd output
root@fffff:/app/graphrag-folder/output# ls
communities.parquet        embeddings.community.full_content.parquet  graph.graphml          text_units.parquet
community_reports.parquet  embeddings.entity.description.parquet      lancedb
context.json               embeddings.text_unit.text.parquet          relationships.parquet
documents.parquet          entities.parquet                           stats.json
root@fffff:/app/graphrag-folder/output# cd lancedb
root@fffff:/app/graphrag-folder/output/lancedb# ls
default-community-full_content.lance  default-entity-description.lance  default-text_unit-text.lance
```

Step 5 – write index output to DB

This step stores the graph output data in DB, as docker backup.
The content in /app/graphrag-folder/output will be saved into DB. This example uses Azure SQL database as DB.

```
$ docker compose up graphrag-writer
```

```
[+] Running 4/4
 ✓ Container postgres             Running                                                    0.0s
 ✓ Container load-data-app        Created                                                    0.0s
 ✓ Container graphrag-index-app   Created                                                    0.0s
 ✓ Container graphrag-writer-app  Recreated                                                  0.2s
Attaching to graphrag-writer-app
graphrag-writer-app  | All relevant files have been processed and stored in Azure SQL .
graphrag-writer-app exited with code 0
```

Step 6 – build graph, prepare for Cypher query

This step builds graph using AGE on PostgreSQL in docker, to prepare for Cypher query.

```
$ docker compose up build-graph
```

```
[+] Running 5/5
 ✓ Container postgres            Running                                    0.0s
 ✓ Container load-data-app       Created                                    0.0s
 ✓ Container graphrag-index-app  Created                                    0.0s
 ✓ Container graphrag-writer-app Created                                    0.0s
 ✓ Container build-graph-app     Recreated                                  0.1s
Attaching to build-graph-app
build-graph-app  | Existing graph 'graphRAG' dropped.
build-graph-app  | Graph 'graphRAG' will be created.
build-graph-app  | Inserted 10 rows into Document
build-graph-app  | Index(['id', 'human_readable_id', 'title', 'text', 'text_unit_ids',
build-graph-app  |        'creation_date', 'metadata'],
build-graph-app  |       dtype='object')
build-graph-app  | Inserted 437 rows into Entity
build-graph-app  | Index(['id', 'human_readable_id', 'title', 'type', 'description',
build-graph-app  |        'text_unit_ids', 'frequency', 'degree', 'x', 'y'],
build-graph-app  |       dtype='object')
build-graph-app  | Inserted 696 rows into relationship
build-graph-app  | Index(['id', 'human_readable_id', 'source', 'target', 'description', 'weight',
build-graph-app  |        'combined_degree', 'text_unit_ids'],
build-graph-app  |       dtype='object')
build-graph-app  | Data loaded into AGE graph successfully.
build-graph-app exited with code 0
```

View the graph content

This step verifies graph in PostgreSQL ag_catalog in docker, name graphRAG.

```
$ docker exec -it postgres bash
```

```
root@fffff:/app# psql -U postgres -d postgres
psql (16.9 (Debian 16.9-1.pgdg120+1))
Type "help" for help.

postgres=# LOAD 'age';
LOAD
postgres=# SET search_path = ag_catalog, "$user", public;
SET
postgres=# SELECT * FROM cypher('graphRAG', $$
    MATCH ()--() RETURN count(*)
$$) AS (count agtype);
 count
-------
 1392
(1 row)

postgres=# SELECT * FROM cypher('graphRAG', $$
    MATCH ()-[r]->() RETURN count(r)
$$) AS (count agtype);
 count
-------
 696
(1 row)
```

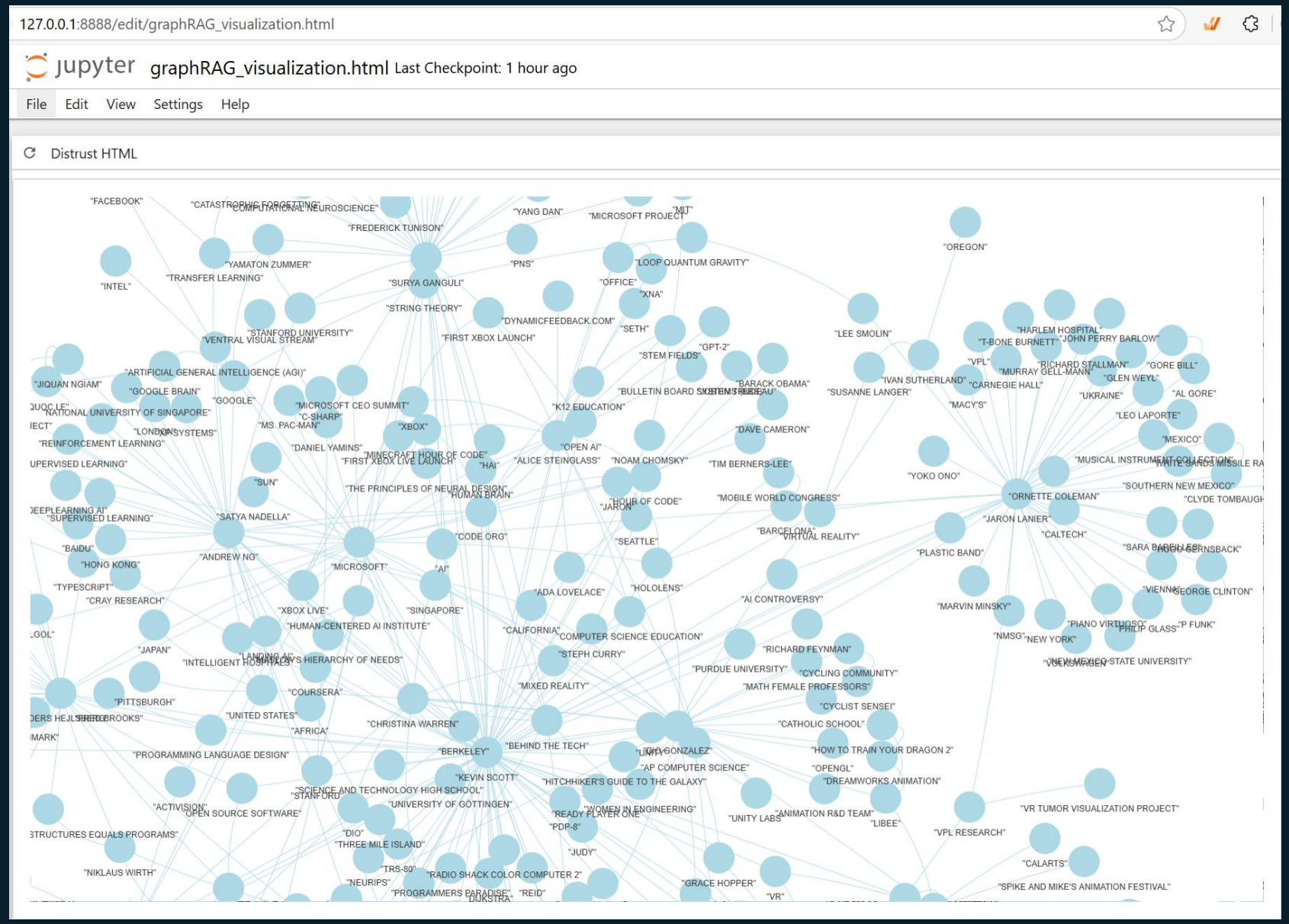
```
root@fffff:/app# psql -U postgres -d postgres
psql (16.9 (Debian 16.9-1.pgdg120+1))
Type "help" for help.

postgres=# LOAD 'age';
LOAD
postgres=# SET search_path = ag_catalog, "$user", public;
SET
postgres=# \dt
              List of relations
   Schema   |   Name   | Type  |  Owner
------------+----------+-------+----------
 ag_catalog | ag_graph | table | postgres
 ag_catalog | ag_label | table | postgres
(2 rows)

postgres=# SELECT * FROM ag_catalog.ag_label;
       name        | graph | id | kind |          relation           |        seq_name
------------------+-------+----+------+-----------------------------+------------------------
 _ag_label_vertex | 87167 |  1 | v    | "graphRAG"._ag_label_vertex | _ag_label_vertex_id_seq
 _ag_label_edge   | 87167 |  2 | e    | "graphRAG"._ag_label_edge   | _ag_label_edge_id_seq
 Document         | 87167 |  3 | v    | "graphRAG"."Document"       | Document_id_seq
 Entity           | 87167 |  4 | v    | "graphRAG"."Entity"         | Entity_id_seq
 RELATED_TO       | 87167 |  5 | e    | "graphRAG"."RELATED_TO"     | RELATED_TO_id_seq
(5 rows)


postgres=# SELECT * FROM ag_catalog.ag_graph;
 graphid |   name   | namespace
---------+----------+------------
   87167 | graphRAG | "graphRAG"
(1 row)
```

Graph visualization



Step 7 – run query in jupyter notebook

This step runs jupyter notebook in docker.

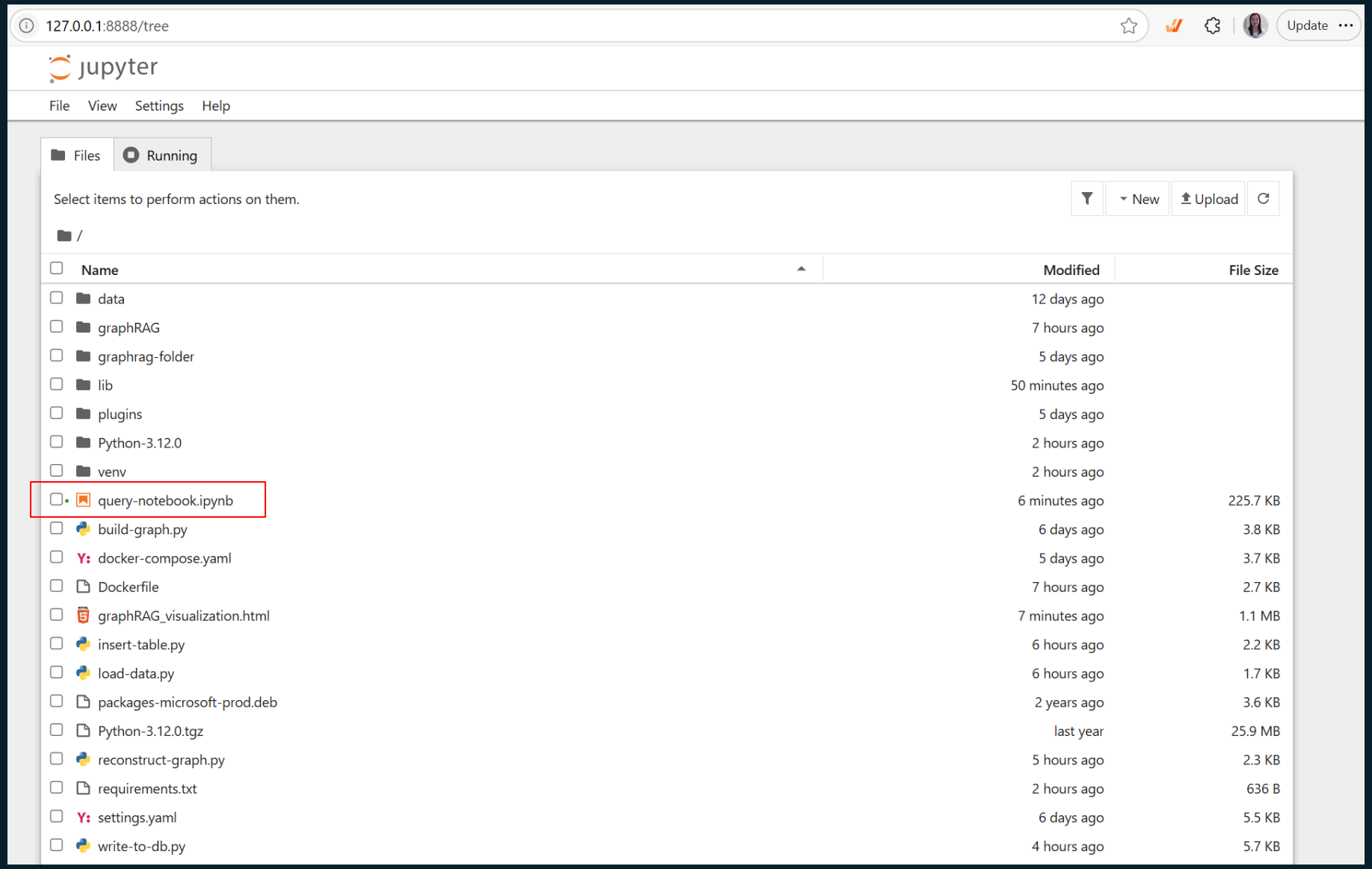
```
$ docker compose up query-notebook
```

```
[+] Running 6/6
 ✓ Container postgres            Running                                    0.0s
 ✓ Container load-data-app       Created                                    0.0s
 ✓ Container graphrag-index-app  Created                                    0.0s
 ✓ Container graphrag-writer-app Created                                    0.0s
 ✓ Container build-graph-app     Created                                    0.0s
 ✓ Container query-notebook-app  Recreated                                  0.1s
Attaching to query-notebook-app
query-notebook-app  | [I 2025-05-23 16:51:30.721 ServerApp] jupyter_lsp | extension was successfully linked.
query-notebook-app  | [I 2025-05-23 16:51:30.727 ServerApp] jupyter_server_terminals | extension was successfully linked.
query-notebook-app  | [W 2025-05-23 16:51:30.728 LabApp] 'shutdown_no_activity_timeout' has moved from NotebookApp to ServerApp. Th
is config will be passed to ServerApp. Be sure to update your config before our next release.
query-notebook-app  | [I 2025-05-23 16:51:30.730 ServerApp] jupyterlab | extension was successfully linked.
query-notebook-app  | [I 2025-05-23 16:51:30.734 ServerApp] notebook | extension was successfully linked.
query-notebook-app  | [I 2025-05-23 16:51:30.742 ServerApp] Writing Jupyter server cookie secret to /root/.local/share/jupyter/runt
ime/jupyter_cookie_secret
query-notebook-app  | [I 2025-05-23 16:51:31.018 ServerApp] notebook_shim | extension was successfully linked.
query-notebook-app  | [I 2025-05-23 16:51:31.060 ServerApp] notebook_shim | extension was successfully loaded.
query-notebook-app  | [I 2025-05-23 16:51:31.062 ServerApp] jupyter_lsp | extension was successfully loaded.
query-notebook-app  | [I 2025-05-23 16:51:31.063 ServerApp] jupyter_server_terminals | extension was successfully loaded.
query-notebook-app  | [I 2025-05-23 16:51:31.078 LabApp] JupyterLab extension loaded from /app/venv/lib/python3.12/site-packages/ju
pyterlab
query-notebook-app  | [I 2025-05-23 16:51:31.078 LabApp] JupyterLab application directory is /app/venv/share/jupyter/lab
query-notebook-app  | [I 2025-05-23 16:51:31.080 LabApp] Extension Manager is 'pypi'.
query-notebook-app  | [I 2025-05-23 16:51:31.136 ServerApp] jupyterlab | extension was successfully loaded.
query-notebook-app  | [I 2025-05-23 16:51:31.138 ServerApp] notebook | extension was successfully loaded.
query-notebook-app  | [I 2025-05-23 16:51:31.139 ServerApp] Serving notebooks from local directory: /app
query-notebook-app  | [I 2025-05-23 16:51:31.139 ServerApp] Jupyter Server 2.16.0 is running at:
query-notebook-app  | [I 2025-05-23 16:51:31.139 ServerApp] http://

query-notebook-app  | [I 2025-05-23 16:51:31.139 ServerApp]     http://127.0.0.1:8888/tree?token=

query-notebook-app  | [I 2025-05-23 16:51:31.139 ServerApp] Use Control-C to stop this server and shut down all kernels (twice to s
kip confirmation).
query-notebook-app  | [C 2025-05-23 16:51:31.150 ServerApp]
```

Jupyter notebook UI



Cypher query examples

Several ways of querying are provided:

- Cypher query
  - Example given to check nodes and relationships, and their properties.
- Multi-hop cypher query
  - One of the advantages of using knowledge graph is the multi-hop query.
- GraphRAG LocalSearch
  - Examples given to show the impact of community level to the query results.
- GraphRAG GlobalSearch
  - Both python function call and CLI methods are provided.
- Vector search
  - Example shows that vector search result is restricted to the original text; it typically returns the top-k most similar original chunks. In other words, vector search is retrieval only, it returns what was said, not what it means. However, graphRAG combines vector search with a knowledge graph that encodes relationships between entities. The graph is built from LLM-extracted triples, which helps contextual thinking; thus, graphRAG provides more synthesized and thoughtful response.

Two vector search methods

Use AOAI in local postgres, without pgvector

Use pgvector extension in local postgres.

It creates a table graphrag_vectors in postgres

```
postgres=# \dt
                List of relations
   Schema   |       Name       | Type  |  Owner
------------+------------------+-------+----------
 ag_catalog | ag_graph         | table | postgres
 ag_catalog | ag_label         | table | postgres
 public     | graphrag_vectors | table | postgres
(3 rows)

postgres=#
SELECT COUNT(*) FROM graphrag_vectors;
 count
-------
   119
(1 row)

postgres=#
SELECT * FROM graphrag_vectors LIMIT 5;
postgres=# \dx
                              List of installed extensions
  Name   | Version |   Schema   |                     Description
---------+---------+------------+-----------------------------------------------------
 age     | 1.5.0   | ag_catalog | AGE database extension
 plpgsql | 1.0     | pg_catalog | PL/pgSQL procedural language
 vector  | 0.8.0   | public     | vector data type and ivfflat and hnsw access methods
(3 rows)
```

Build and run AI agents example1

Task: summarize the podcasts.

```
# Now call your function
await summarize_all_text("/app/graphrag-folder/input", agent)
```

```
✔ Loaded: input_8.txt
✔ Loaded: input_3.txt
✔ Loaded: input_7.txt
✔ Loaded: input_4.txt
✔ Loaded: input_6.txt
✔ Loaded: input_9.txt
✔ Loaded: input_1.txt
✔ Loaded: input_10.txt
✔ Loaded: input_2.txt
✔ Loaded: input_5.txt

--- Combined Summary ---
 Here’s a concise summary of the text:

In the Behind the Tech Podcast hosted by Kevin Scott, Microsoft’s CTO, Scott engages in conversations with eminent individuals in technolog
y, delving into their journeys, fascinating stories, and impactful innovations. The highlighted guests include Jaron Lanier discussing his c
oncerns about social media's negative influence, and the internet's trajectory; Reid Hoffman exploring entrepreneurial strategies like blitz
scaling; Danielle Feinberg reflecting on creativity and coding at Pixar; and Alice Steinglass advocating for inclusive computer science educ
ation in K-12 through Code.org. Guests enrich the narrative with diverse personal paths into technology and share actionable advice for leve
raging coding, innovation, and interdisciplinary collaboration. They highlight challenges in AI development, diversity, and empowering commu
nities through tech, fostering broader engagement in the tech revolution and inspiring future generations.

"Here’s a concise summary of the text:\n\nIn the Behind the Tech Podcast hosted by Kevin Scott, Microsoft’s CTO, Scott engages in conversati
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logy and share actionable advice for leveraging coding, innovation, and interdisciplinary collaboration. They highlight challenges in AI dev
elopment, diversity, and empowering communities through tech, fostering broader engagement in the tech revolution and inspiring future gener
ations."
```

Build and run AI agents example2

Task: convert natural language query to Cypher query.

```python
from semantic_kernel.functions.kernel_arguments import KernelArguments

# Replace this with any natural language question
user_input = "Who did KEVIN SCOTT mention that is connected to computer science, education, or kids?"

# Create arguments
arguments = KernelArguments(user_input=user_input)

# Get the registered function
nl2cypher_function = kernel.get_function("GraphRAG", "NL2CypherGraphRAG")

# Invoke the function
result = await nl2cypher_function.invoke(kernel, arguments)

# Print the generated Cypher query
print("🧠 Generated Cypher Query:\n", result)
```

🧠 Generated Cypher Query:

```sql
SELECT * FROM cypher('graphRAG', $$
    MATCH (kevin:Entity {title: 'KEVIN SCOTT'})-[:RELATED_TO]->(mentioned_person:Entity)-[r:RELATED_TO]->(connected_entity:Entity)
    WHERE toLower(r.description) CONTAINS 'computer science'
       OR toLower(r.description) CONTAINS 'education'
       OR toLower(r.description) CONTAINS 'kids'
    RETURN mentioned_person.name AS mentioned_person,
           connected_entity.name AS connected_entity,
           r.description AS relationship_description
$$) AS (mentioned_person TEXT, connected_entity TEXT, relationship_description TEXT);
```

Optional step: reconstruct-graph

In case the graph related data in docker /app/graphrag-folder/output lost, then the graph can be reconstructed using the data stored in DB, which was saved in step 5.

```
$ docker compose up reconstruct-graph
```

```
[+] Running 2/2
 ✓ Container postgres                Running                                                    0.0s
 ✓ Container reconstruct-graph-app   Created                                                    0.3s
Attaching to reconstruct-graph-app
reconstruct-graph-app  | All files have been reconstructed from Azure SQL to /app/graphrag-folder/restore.
reconstruct-graph-app exited with code 0
```

```
$ docker exec -it postgres bash
```

```
root@fffff:/app/graphrag-folder# cd restore
root@fffff:/app/graphrag-folder/restore# ls
communities.parquet        embeddings.community.full_content.parquet  graph.graphml          text_units.parquet
community_reports.parquet  embeddings.entity.description.parquet      relationships.parquet
context.json               embeddings.text_unit.text.parquet          stats.json
documents.parquet          entities.parquet                           summaries.json
```

Query in docker postges

```
$ docker exec -it postgres bash
```

```
root@fffff:/app# psql -U postgres -d postgres
psql (16.9 (Debian 16.9-1.pgdg120+1))
Type "help" for help.

postgres=# LOAD 'age';
LOAD
postgres=# SET search_path = ag_catalog, "$user", public;
SET
postgres=# SELECT * FROM cypher('graphRAG', $$
    MATCH ()--() RETURN count(*)
$$) AS (count agtype);
 count
-------
 1392
(1 row)

postgres=# SELECT * FROM cypher('graphRAG', $$
    MATCH ()-[r]->() RETURN count(r)
$$) AS (count agtype);
 count
-------
 696
(1 row)
```

Graph storage

Graph metadata, like nodes, edges stored in Docker postgres

```
root@fffff:/app# psql -U postgres -d postgres
psql (16.9 (Debian 16.9-1.pgdg120+1))
Type "help" for help.

postgres=# LOAD 'age';
LOAD
postgres=# SET search_path = ag_catalog, "$user", public;
SET
postgres=# SELECT * FROM cypher('graphRAG', $$
    MATCH ()--() RETURN count(*)
$$) AS (count agtype);
 count
-------
 1392
(1 row)

postgres=# SELECT * FROM cypher('graphRAG', $$
    MATCH ()-[r]->() RETURN count(r)
$$) AS (count agtype);
 count
-------
 696
(1 row)
```

Graph input and output stored in DB

Example: DB is Azure hosted PostgreSQL

```
postgres=> \dt
                List of relations
 Schema |       Name        | Type  |   Owner
--------+-------------------+-------+----------
 public | graphrag_inputs   | table | XXXX
 public | graphrag_outputs  | table | XXXX
(2 rows)
```

```
$ docker exec -it postgres bash
```

```
root@fffff:/app# psql -U postgres -d postgres
psql (16.9 (Debian 16.9-1.pgdg120+1))
Type "help" for help.

postgres=# LOAD 'age';
LOAD
postgres=# SET search_path = ag_catalog, "$user", public;
SET
postgres=# \dt
                List of relations
   Schema   |   Name    | Type  |  Owner
------------+-----------+-------+----------
 ag_catalog | ag_graph  | table | postgres
 ag_catalog | ag_label  | table | postgres
(2 rows)

postgres=# SELECT * FROM ag_catalog.ag_label;
       name        | graph | id | kind |          relation           |        seq_name
-------------------+-------+----+------+-----------------------------+------------------------
 _ag_label_vertex  | 87167 |  1 | v    | "graphRAG"._ag_label_vertex | _ag_label_vertex_id_seq
 _ag_label_edge    | 87167 |  2 | e    | "graphRAG"._ag_label_edge   | _ag_label_edge_id_seq
 Document          | 87167 |  3 | v    | "graphRAG"."Document"       | Document_id_seq
 Entity            | 87167 |  4 | v    | "graphRAG"."Entity"         | Entity_id_seq
 RELATED_TO        | 87167 |  5 | e    | "graphRAG"."RELATED_TO"     | RELATED_TO_id_seq
(5 rows)


postgres=# SELECT * FROM ag_catalog.ag_graph;
 graphid |   name    | namespace
---------+-----------+------------
   87167 | graphRAG  | "graphRAG"
(1 row)
```

Example: DB is Azure SQL database

Run Cancel Disconnect Change Database:

```
1  SELECT TABLE_SCHEMA, TABLE_NAME
2  FROM INFORMATION_SCHEMA.TABLES
3  WHERE TABLE_TYPE = 'BASE TABLE';
4
```

Results Messages

	TABLE_SCHEMA	TABLE_NAME
1	dbo	graphrag_inputs
2	dbo	graphrag_outputs

What’s next?

- Plugin extensibility: The Docker image includes a dedicated /app/plugins directory, designed to support future integrations—such as adding https://Github.com/microsoft/semantic-kernel plugins for orchestration and reasoning.

- AI agent development: Expand the system by developing additional AI agents tailored to specific tasks, domains, or workflows. Multi-agents.

- Scaling options. Containerapp possible.

- Experiment more DB types where the graphRAG input resides. So far, this solution provided examples of Azure SQL database and Azure hosted PostgreSQL as DB. If you have other types of DB, you may need to add relevant python modules in Dockerfile and requirements.txt, and make some adjustments in .env, insert-table.py, load-data.py, write-to-db.py and reconstruct-graph.py.